Luca Buiarelli

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Profile

Condensed matter physicist with experience in analytical modeling, numerical simulations, and group theory. Expertise includes Landau theories, tight-binding models, density functional theory, and Monte Carlo methods, with a current research focus on altermagnetism and unconventional magnetic phases. I developed `ProDenCeR`, an open-source Python tool for post-processing density functional theory calculations to diagnose unconventional electronic and magnetic phases in materials.

Education

University of Minnesota, USA — PhD, Materials Science *Sep. 2023 – Present*

University of Copenhagen, Denmark — MSc, Condensed Matter Physics *Sep. 2021 – May 2023*

University of Pisa, Italy — BSc, Physics *Sep. 2018 – Sep. 2021*

Awards and Fellowships

Matthew Tirrell Graduate Fellowship *2024*

Teaching

Teaching Assistant *Spring and Fall 2024*
University of Minnesota, Introduction to Materials Science Laboratory (MATS 2001/2002)

Service and Outreach

Journal Refereeing
Nature *npj Quantum Materials*, Nature *Communications Materials*

Penne Amiche della Scienza *Italy*
Science outreach program for Italian middle school students involving monthly letter exchanges, aimed at fostering interest in science and academic careers.

Workshops and Conferences

Aspen Center for Physics, Winter Conference on Altermagnetism, Aspen, CO (poster) *April 2026*

American Physical Society March Meeting, Denver, CO (contributed talk) *March 2026*

Flatiron Institute, TRIQS School, Paris, France (attended) *September 2025*

American Physical Society March Meeting, Los Angeles, CA (contributed talk) *March 2025*

School on Electron–Phonon Physics, University of Texas at Austin (attended) *June 2024*

American Physical Society March Meeting, Minneapolis, MN (contributed talk) *March 2024*

Skills

Coding: MATLAB, Python, C
Scientific Communication: Experience preparing scientific manuscripts using LaTeXand delivering oral presentations with PowerPoint slides at international conferences.
Languages: English (fluent), Italian (native), Spanish (basic)

Articles

Published

1. Probing multipolar order in the candidate altermagnet MnF_2 through the elastocaloric effect under strain
Rahel Ohlendorf, **Luca Buiarelli**, Hilary ML Noad, Andrew P Mackenzie, Rafael M Fernandes, Turan Birol, Jörg Schmalian, Elena Gati
Phys. Rev. Lett. **137** (2026) *Editors' Suggestion* [Journal] [arXiv]

2. Observation of mirror-odd and mirror-even spin texture in ultra-thin epitaxially-strained RuO_2 films
Yichen Zhang, Seung Gyo Jeong, **Luca Buiarelli**, Seungjun Lee, Yucheng Guo, Jiaqin Wen, Hang Li, Sreejith Nair, In Hyeok Choi, Zheng Ren, Ziqin Yue, Alexei Fedorov, Sung-Kwan Mo, Junichiro Kono, Jong Seok Lee, Tony Low, Turan Birol, Rafael M. Fernandes, Milan Radovic, Bharat Jalan, Ming Yi
Science Adv. (2026) [arXiv]

3. Electronic and structural properties of Rh- and Pd-based kagome layered shandites from first principles
Luca Buiarelli, Turan Birol, Brian M. Andersen, Morten H. Christensen
Phys. Rev. B **113** (2026) *Editors' Suggestion* [Journal] [arXiv]

4. Altermagnetic Polar Metallic Phase in Ultra-Thin Epitaxially-Strained RuO_2 Films
Seung Gyo Jeong, In Hyeok Choi, Sreejith Nair, **Luca Buiarelli**, Bita Pourbahari, Jinyoung Oh, Nabil Bassim, Ambrose Seo, Woo Seok Choi, Rafael M. Fernandes, Turan Birol, Liuyan Zhao, Jong Seok Lee, Bharat Jalan
Proc. Natl. Acad. Sci. **123** (2026)[Journal] [arXiv]

5. Noncollinear Magnetic Multipoles in Collinear Altermagnets
Luca Buiarelli, Rafael M. Fernandes, Turan Birol
Phys. Rev. B **112**, 224442 (2025) [Journal] [arXiv]

Preprints

1. Discovery of an odd-parity f-wave charge order in a kagome metal
Jiangchang Zheng, Caiyun Chen, Ruiqin Fu, **Luca Buiarelli**, Zihan Lin, Fazhi Yang, Tianhao Guo, Ganesh Pokharel, Andrea Capa Salinas, Sen Zhou, Turan Birol, Stephen D. Wilson, Junzhang Ma, Daniel J. Schultz, Xianxin Wu, Berthold Jäck
Submitted [arXiv]

2. Interplay between charge correlations and superconductivity across the superconducting domes of $CsV_3Sb_{5-x}Sn_x$
Andrea N. Capa Salinas, Brenden R. Ortiz, Steven J. Gomez Alvarado, Sarah Schwarz, Ganesh Pokharel, **Luca Buiarelli**, Hyeonseo Harry Park, Shiyu Yuan, Roland Yin, Suchismita Sarker, Turan Birol, Stephen D. Wilson
Submitted [arXiv]